Unsupervised Machine Learning

Machine learning 2

Muhammed Elameiry

Senior Data Scientist, Banque Misr

Anomaly Detection

Today’s Agenda

- Anomaly Detection

  Local Outlier Factor (LOF)
  Isolation Forest

Local Outlier Factor (LOF)

Local Outlier Factor (LOF) algorithm computes a score (called local outlier factor) reflecting the degree of abnormality of the observations. It measures the local density deviation of a given data point with respect to its neighbors. The idea is to detect the samples that have a substantially lower density than their neighbors.

What problem LOF solves

LOF addresses the failure of global distance-based methods in datasets where **normal behavior occurs at different densities** across the feature space.

Type of anomalies detected

- **Local anomalies**: points that are normal globally but abnormal relative to nearby points
- Not designed for purely **global isolation**
- Particularly effective in **varying-density datasets**

Why Distance Alone Is Not Enough

Distance-based anomaly detection assumption
Points that are far from most other points are anomalies.

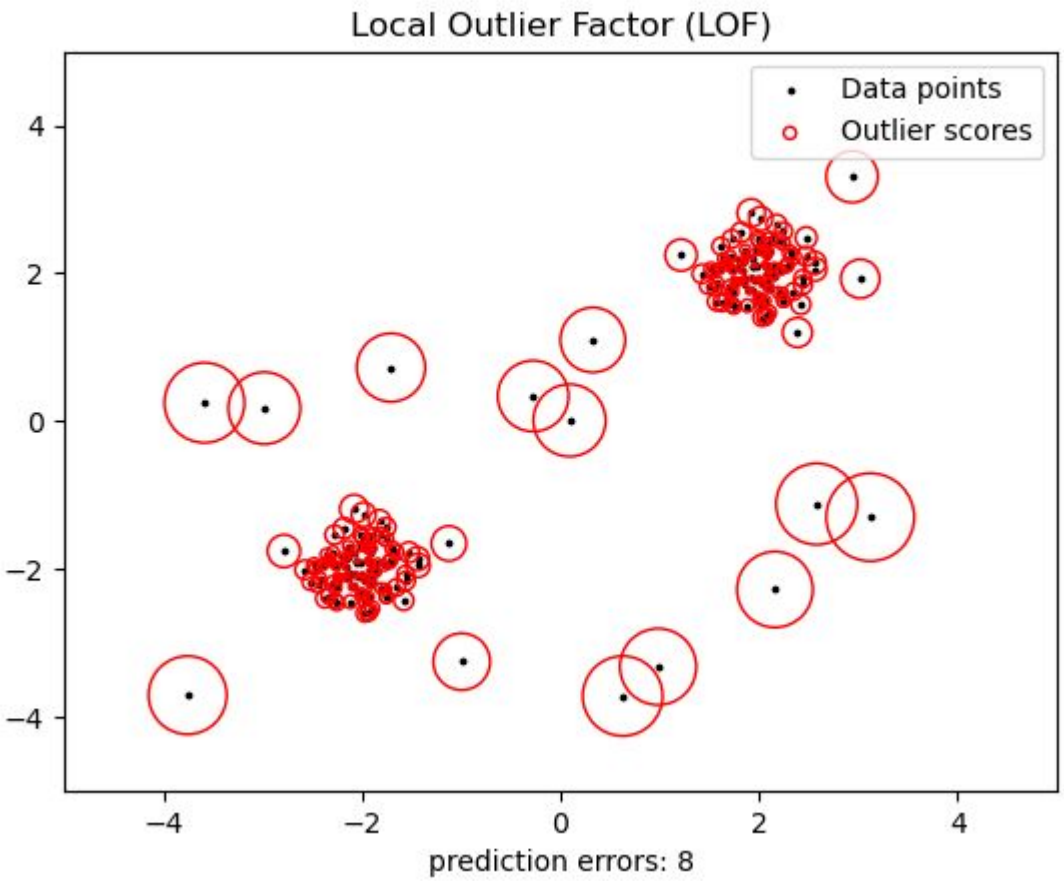
Why this fails
- Real datasets often contain:
- Dense clusters
- Sparse clusters
- Gradual density changes

A point in a sparse cluster may be far from others but still **locally normal**

Core limitation
Global distance ignores **local context**.

Key contrast
- Global distance: *How far am I from everyone?*
- Local density: *How crowded am I compared to my neighbors?*

LOF adopts the second perspective.


Local Outlier Factor (LOF)
Data points
Outlier scores
4
2
0
−2
−4
−4
−2
0
2
4
prediction errors: 8

Step-by-Step LOF Algorithm

For each data point x:

Step 1 — k-Nearest Neighbors

Identify the set $N_k(x)$

Step 2 — k-distance

•Compute the distance from x to its k-th nearest neighbor

Step 3 — Reachability Distance

•Stabilize distances using neighbors' local scale

Step 4 — Local Reachability Density (LRD)

•Convert reachability distances into a local density estimate

Step 5 — LOF Score

•Compare the density of x to the average density of its neighbors

Definitions

$$k\text{-distance}(x)$$

Distance from x to its k-th nearest neighbor.

$$N_k(x)$$

Set of k-nearest neighbors of x.

$$\text{reach-dist}_k(x, o)$$

Reachability distance between x and neighbor o.

$$\text{LRD}_k(x)$$

Local reachability density of x.

$$\text{LOF}_k(x)$$

Local Outlier Factor score of x.

What is $k-distance(\cdot)$?

Precise definition

$$k\text{-distance}(x) = \text{distance from } x \text{ to its k-th nearest neighbor}$$

Important properties

- Each point has its **own** k-distance
- $k\text{-distance}(x) \neq k\text{-distance}(o)$ in general

Why this matters

- Different regions have different density scales
- k-distance captures the **local neighborhood radius** around each point

Interpretation

> $k\text{-distance}(x)$ describes how far one must go from x to see k nearby points.

What is Reachability Distance?

Definition

$$\text{reach-dist}_k(x, o) = \max\left(k\text{-distance}(o),\ d(x, o)\right)$$

What it is NOT

- Not subtraction
- Not a new metric
- Not asymmetric by accident (asymmetry is intentional)

What it IS

- A **lower-bounded distance**
- Distance from x to o, floored by the neighborhood scale of o

Key dependency

Reachability distance depends on **the neighbor's neighborhood**, not just on x.

Why We Take the MAX in Reachability Distance

Case 1: $d(x, o) < k\text{-}distance(o)$

$$\text{reach-dist}_k(x, o) = k\text{-}distance(o)$$

Meaning

- Even if x is very close to o, we treat the distance as at least the typical spacing in o's neighborhood.

Why

- Prevents artificially small distances
- Avoids unrealistically high density estimates

Case 2: $d(x, o) > k\text{-}distance(o)$

$$\text{reach-dist}_k(x, o) = d(x, o)$$

Meaning

- True separation is preserved
- No smoothing when points are genuinely far apart

Overall effect

- Stabilizes density estimation
- Makes LOF robust across regions with different densities

Local Reachability Density (LRD)

Definition

$$\mathrm{LRD}_k(x) = \left(\frac{1}{|N_k(x)|} \sum_{o \in N_k(x)} \text{reach-dist}_k(x, o) \right)^{-1}$$

Interpretation

- Average reachability distance to neighbors
- Inverse converts distance into **density**

Meaning

- High LRD → dense neighborhood
- Low LRD → sparse neighborhood

Density here is **purely local**.

What is Reachability Distance?

Definition

$$\mathrm{LOF}_k(x) = \frac{\frac{1}{|N_k(x)|} \sum_{o \in N_k(x)} \mathrm{LRD}_k(o)}{\mathrm{LRD}_k(x)}$$

Interpretation

- Ratio of neighbors' average density to point's density
- Measures **local density deviation**

Expected behavior

- LOF ≈ 1 → density similar to neighbors → normal
- LOF > 1 → neighbors denser → potential outlier

Worked Numerical Example

Assume $k = 3$.

For point A:

Neighbor distances and neighborhood scales

Neighbor o	$d(A, o)$	k-distance(o)
B	2	4
C	3	3
D	5	6

Worked Numerical Example

Reachability distances

$$\text{reach-dist}(A, B) = \max(4, 2) = 4$$

$$\text{reach-dist}(A, C) = \max(3, 3) = 3$$

$$\text{reach-dist}(A, D) = \max(6, 5) = 6$$

Average reach-dist:

$$\frac{4 + 3 + 6}{3} = 4.33$$

Worked Numerical Example

Local Reachability Density

$$\mathrm{LRD}_3(A) = \frac{1}{4.33} \approx 0.23$$

Assume:

- $\mathrm{LRD}(B) = 0.50$
- $\mathrm{LRD}(C) = 0.45$
- $\mathrm{LRD}(D) = 0.40$

Average neighbor LRD:

$$\frac{0.50 + 0.45 + 0.40}{3} = 0.45$$

Worked Numerical Example

LOF score

$$\mathrm{LOF}_3(A) = \frac{0.45}{0.23} \approx 1.96$$

Conclusion

Point A is significantly less dense than its neighbors → **local outlier**.

LOF ≈ 1
Density comparable to neighbors → normal

LOF > 1
Local density lower than neighbors → anomalous

Isolation forest

Isolation forest

Definition
Isolation Forest is an unsupervised anomaly detection algorithm that identifies anomalies by **isolating observations using random splits**. Points that are isolated with **fewer splits** are considered anomalous.

Problem it solves
Detects anomalies **without computing distances or densities**, avoiding common failures in complex or high-dimensional data.

Why it works well in high dimensions
It relies on **random partitioning**, not geometry. Distances and densities degrade in high dimensions; random splits do not.

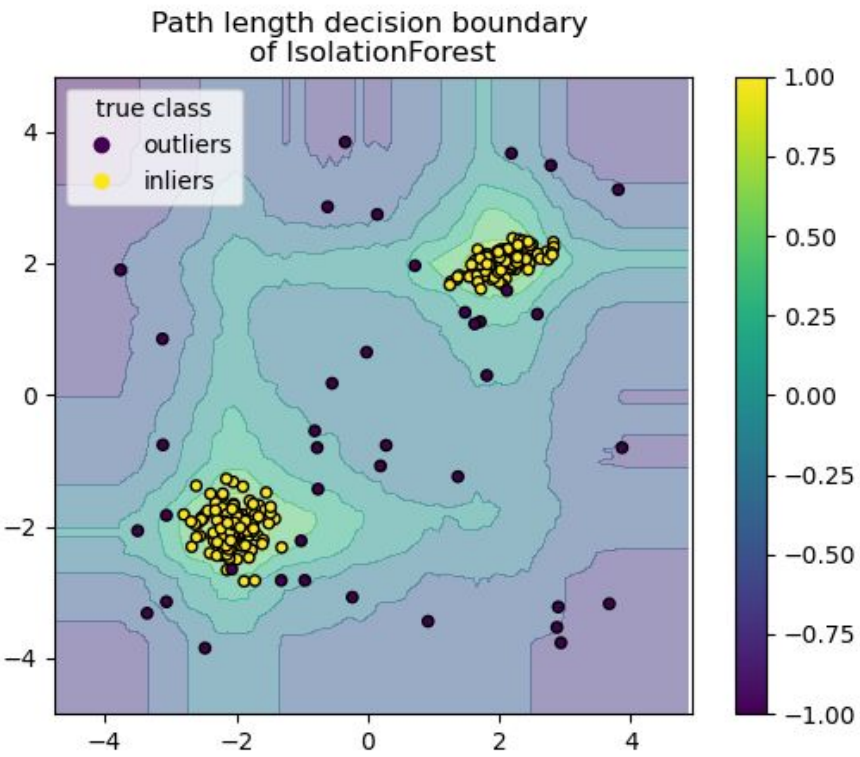
Isolation forest

Traditional anomaly detection:
- Measures how far a point is
- Or how dense its neighborhood is

Isolation Forest:
- Does **not** model normal behavior
- Does **not** define neighborhoods
- Does **not** estimate probabilities

Core idea
- Anomalies are easier to isolate than norm points.

Normal points are surrounded by many other
Anomalies are few and lie in sparse regions.



Isolation forest

Isolation Forest builds trees using:

- Random feature selection
- Random split values
- Recursive partitioning

Each split divides the data into two parts.

Why randomness helps

- No assumptions about data shape
- No optimization objective
- Quickly separates rare points

Randomness makes the algorithm **simple, fast, and robust**.

Why Anomalies Require Fewer Splits

Dense regions:

- Many points clustered together
- Many splits needed to separate one point

Sparse regions:

- Few points
- One or two splits often enough

Key observation

- Short path → easy to isolate → anomaly
- Long path → hard to isolate → normal

This property holds across dimensions.

Isolation forest - Algorithm Overview

Step 1: Subsampling the data

Randomly sample $n$points from the dataset (without replacement).

Step 2: Random feature selection

At each node, randomly choose one feature.

Step 3: Random split value

Select a split uniformly between the feature's minimum and maximum.

Step 4: Recursive partitioning

Split data and repeat until:
one point remains, or
maximum depth is reached

Step 5: Path length computation

Record the depth at which each point is isolated.

Step 6: Averaging across trees

Repeat steps 1–5 across many trees and average path lengths.

Step 7: Anomaly score computation

Convert average path length into a normalized anomaly score.

Isolation forest

Let:

- x: data point
- $h(x)$: path length of x in a single tree
- $E[h(x)]$: expected path length across trees
- n: subsample size
- $c(n)$: normalization constant

Path length

$$h(x) = \text{number of splits required to isolate } x$$

Expected path length

$$E[h(x)] = \text{average of } h(x) \text{ over all trees}$$

Normalization Constant $c(n)$:Expected Path Length

Definition

The normalization constant is defined as:

$$c(n) = 2H(n-1) - \frac{2(n-1)}{n}$$

where $H(\cdot)$ is the **harmonic number**.

Why path length depends on sample size

- Larger samples contain more points
- More splits are required **on average** to isolate a typical point
- Raw path lengths **increase with** n

Therefore, raw path length values are **not comparable** across different subsample sizes.

Anomaly Score Function

Definition

The anomaly score for a point x is defined as:

$$s(x) = 2^{-\frac{E[h(x)]}{c(n)}}$$

where:

- $E[h(x)]$ is the average path length of x across trees
- $c(n)$ is the expected path length of a normal point for subsample size n

Properties

- The score is **bounded in (0, 1]**
- **Shorter path length → higher score**
- **Longer path length → lower score**

Anomaly Score Interpretation

- $s(x) \approx 1$

  Very short path → strong anomaly
- $s(x) \approx 0.5$

  Path length similar to average → ambiguous
- $s(x) \ll 0.5$

  Long path → normal point

Relationship

- Short path length $\Longleftrightarrow$ high anomaly likelihood
- Long path length $\Longleftrightarrow$ normal behavior